



CE351: TERM PROJECT

Determining the Mode Distribution Percentages by Using Four-Step Travel Demand Modelling Framework

PREPARED BY

Md. Abeer Azad Niloy

Id: 2023-3-22-007

SUPERVISED BY

Dr. Fuad Yasin Huda (FYH)

Senior Lecturer

**Department of Civil Engineering
East West University**

Executive Summary

Traffic congestion is a major concern in the Mirpur area of Dhaka. This study focuses on the Mirpur DOHS–Mirpur 1 corridor, where the reported commuting time is about 55 minutes, although the journey is expected to take around 35 minutes under efficient traffic conditions. The main purpose of this study is to understand the amount of travel in the area, where the trips are going, and which transport modes people are likely to use.

Travel time, travel cost, comfort, safety, and reliability were considered in estimating the likelihood of choosing each mode. Household and land-use information were used to estimate the number of trips produced and attracted by each zone. The analysis estimated a total of 14,207 trips/day. These trips were then distributed between the zones using a mathematical model based on the number of trips in each zone and the travel time between them. Recent travel-time observations were collected from Google Maps during the morning and evening peak periods instead of relying only on the older travel-time data provided in the project.

The study also examined how the estimated trips may be divided among Bicycle/Rickshaw, Public Transit, and Private Vehicle. Travel time, travel cost, comfort, safety, and reliability were considered in estimating the likelihood of choosing each mode. The results predicted that 97.84% of the trips would use Private Vehicles, while 1.65% would use Bicycle/Rickshaw and 0.51% would use Public Transit. For the Mirpur DOHS–Mirpur 1 movement, the bestimated travel demand was 130 trips/day from Mirpur DOHS to Mirpur 1 and 65 trips/day in the reverse direction.

The results indicate a strong dependence on private vehicles in the estimated travel demand. This may be one factor contributing to congestion along the study corridor. However, the result should be interpreted with caution because only 11 respondents were included in the mode-choice survey. In addition, the travel time, cost, comfort, safety, and reliability values were mainly collected for the Mirpur 1–Mirpur 10 movement and may not fully represent other trips in the study area.

Overall, the study provides an initial picture of travel demand and transport mode use in the selected Mirpur area. Improving public transport service, reducing unnecessary dependence on private vehicles, collecting more survey data, and further investigating road and intersection conditions are recommended for future study.

Contents

Executive Summary	i
1 Introduction	1
2 Methodology and Data Evaluation	1
2.1 Methodology	1
2.2 Data Evaluation	2
3 Model Setup and Detailed Calculations	3
3.1 Task 1: Trip Generation	3
3.1.1 Trip Production	3
3.1.2 Trip Attraction	3
3.2 Task 2: Trip Distribution	6
3.2.1 Selection of Gravity Model	6
3.2.2 Travel Time Data Collection	6
3.2.3 Determination of Friction Factors	7
3.2.4 Initial Gravity Model Calculation	8
3.2.5 Doubly Constrained Gravity Model	10
3.3 Task 3: Mode Choice Analysis	13
3.3.1 Discrete Choice Modelling	13
3.3.2 Multinomial Logit Model	14
3.3.3 Maximum Likelihood Estimation	14
3.3.4 Specification of Utility Functions	15
3.3.5 Literature-Based Coefficients	15
3.3.6 Survey Data Collection	16
3.3.7 Limitations of the Survey and Mode Choice Model	17
3.3.8 MLE Calibration of Unknown Coefficients	17
3.3.9 MLE Results	19
3.3.10 Final Utility Functions	20
3.3.11 Determination of Utility Value	20
3.3.12 Determination of Mode Choice Probabilities	21
3.3.13 Determination of Mode-Specific Trips	22
4 Results and Discussion	23
4.1 Trip Generation and Distribution	23
4.2 Mode Choice and Implications for Congestion	24
5 Conclusion and Recommendations	24
5.1 Conclusion	24
5.2 Recommendations	25
References	26
A Trip Generation and Trip Attraction Data	27
A.1 Trip Production and Trip Attraction Rates	27
A.2 Household Income Data	27
A.3 Land-Use Data	27
A.4 Raw Trip Production and Trip Attraction	28
A.5 Balanced Trip Attraction	28
A.6 Final Balanced Trip Production and Trip Attraction	28

B	Google Maps Travel Time Data	29
B.1	Sunday, 8 AM	29
B.2	Sunday, 5 PM	29
B.3	Monday, 8 AM	29
B.4	Monday, 5 PM	29
B.5	Tuesday, 8 AM	30
B.6	Tuesday, 5 PM	30
B.7	Wednesday, 8 AM	30
B.8	Wednesday, 5 PM	30
B.9	Thursday, 8 AM	31
B.10	Thursday, 5 PM	31
B.11	Sunday, Average of 8 AM and 5 PM	31
B.12	Monday, Average of 8 AM and 5 PM	31
B.13	Tuesday, Average of 8 AM and 5 PM	32
B.14	Wednesday, Average of 8 AM and 5 PM	32
B.15	Thursday, Average of 8 AM and 5 PM	32
B.16	Final, Average of 5 Days	32
C	Friction Factor Calculation	33
C.1	Project-Provided Friction Factors	33
C.2	Interpolated Friction Factors	33
D	Gravity Model Calculation	34
D.1	Initial Gravity Model Calculation	34
D.2	Second Iteration	36
E	Mode Choice Modelling Data	38
E.1	Survey Responses for Mode Choice Analysis	38
E.2	Detailed MNL and Maximum Likelihood Estimation Calculation	39

List of Tables

1	Raw Trip Production (TP) and Trip Attraction (TA)	4
2	Balanced Trip Attraction	4
3	Final Balanced Trip Production and Trip Attraction	5
4	Representative Average Travel Time Matrix Developed from Google Maps Observations	7
5	Travel Time and Corresponding Friction Factors Adopted in This Study	7
6	Initial Origin–Destination Matrix Obtained from the Singly Constrained Gravity Model .	9
7	Students Who Shared the Calculation Data	9
8	Trip Productions and Adjusted Trip Attractions for the Second Iteration	11
9	Final Origin–Destination Matrix from the Doubly Constrained Gravity Model	12
10	Convergence Check of the Final Doubly Constrained Gravity Model	12
11	Literature-based coefficients adopted for the mode choice model	16
12	Estimated coefficients from maximum likelihood estimation	19
13	Respondent-level MNL and MLE calculation	20
14	Average Mode-Specific Values from the Survey	21
15	Estimated Mode Choice Probabilities	22
16	Mode-Specific Trips Between Mirpur 1 and Mirpur DOHS	22

List of Figures

1	Representative Google Maps Travel Time Observation (Mirpur 1 to Mirpur 10, Sunday, 8:00 AM)	7
2	Hierarchy of mode choice modelling approaches and the method adopted in the present study.	13
3	Setup of Solver Parameter	18
4	Snapshot of Excel Sheet While Solving MLE	19
5	Estimated Mode-Specific Trips Between Mirpur 1 and Mirpur DOHS	23

1 Introduction

Mirpur is a densely populated area of Dhaka North City Corporation (DNCC) with substantial daily travel demand. A key concern in the study area is the congestion experienced along the Mirpur DOHS–Mirpur 1 corridor. The reported average commuting time between these two locations is approximately 55 minutes, although the travel distance is relatively short and the journey is expected to take about 35 minutes under efficient traffic conditions. This difference indicates that traffic congestion and travel demand characteristics may be important factors affecting the corridor’s travel time.

To better understand this problem, a traditional four-step travel demand modelling approach was applied to the study area. The model provides a structured way to estimate how many trips are generated, where those trips are distributed, and how travellers choose between available transport modes. The project focuses on the first three stages of the framework: trip generation, trip distribution, and mode choice analysis [1].

The study area was divided into eight Traffic Analysis Zones (TAZs). Trip generation was estimated using the available household socio-economic and land-use data. For trip distribution, a gravity model was used to develop the O–D matrix based on zonal trip productions, attractions, and travel time. Recent travel-time observations were also collected to avoid relying only on the older travel-time dataset provided in the project. Finally, mode choice was analysed for Bicycle/Rickshaw (BR), Public Transit (PT), and Private Vehicle (PV) using a Multinomial Logit (MNL) model.

The main purpose of this study is therefore to estimate the travel demand pattern and mode share within the study area and to examine what these results suggest about the congestion problem. The results are intended to provide a basic understanding of the existing travel behaviour and support future transport planning for the study area.

2 Methodology and Data Evaluation

2.1 Methodology

This study followed the traditional four-step travel demand modelling forecasting to investigate the travel demand and its behaviour of identified congestion problem, focusing on trip generation, trip distribution, and mode choice. The study area was divided into eight Traffic Analysis Zones (TAZs), and the analysis was carried out to identify the number of trips among these zones.

First, trip generation was performed using the cross-classification method [2]. Household income data were used to estimate trip productions, while land-use data were used to estimate trip attractions. Because initial total productions and attractions naturally diverge, the attraction values were mathematically balanced against the household-based trip production totals before proceeding to trip distribution.

For trip distribution, a gravity model was selected. Growth factor models were deemed unsuitable as no reliable base-year O–D matrix was available for the study area. The balanced trip productions and attractions served as the trip-end constraints, while travel time served as the measure of spatial separation (impedance). A doubly constrained gravity model was applied iteratively [3] until the calculated attractions satisfied the 5% convergence criterion, ensuring both production and attraction totals were respected.

For the mode choice analysis, a discrete choice approach utilizing the Multinomial Logit (MNL) model was adopted. The three alternatives specified in the project brief were Bicycle/Rickshaw (BR), Public Transit (PT), and Private Vehicle (PV). Travel time, travel cost, comfort, safety, and reliability were evaluated as primary mode attributes. The travel-time and travel-cost coefficients, alongside alternative-specific constants, were adopted from an existing Dhaka mode choice study by Rahman et al. [4]. The additional coefficients for comfort, safety, and reliability were calibrated from original survey data via

Maximum Likelihood Estimation (MLE) using the GRG Nonlinear engine within Microsoft Excel Solver. The resulting mode choice probabilities were subsequently applied to the final O–D matrix to obtain mode-specific trip volumes.

2.2 Data Evaluation

The data utilized in this study originated from three primary sources: the project-provided household and land-use data, empirical travel-time observations collected via Google Maps, and a targeted survey conducted specifically for the mode choice calibration.

The available project data were suitable for the cross-classification method, as both household income and land-use parameters were detailed for each TAZ. The balancing procedure successfully normalized the data, yielding a conserved total of 14,207 trips/day across the network.

Addressing concerns regarding the outdated 2015 travel-time dataset provided in the project brief, contemporary travel-time data were collected for all O–D pairs using Google Maps routing. To ensure consistency and avoid anomalies from non-recurring congestion, departure times were fixed at 8:00 AM and 5:00 PM across five consecutive weekdays. The averages were then used as the updated travel-time values in the trip distribution analysis. However, it is acknowledged that these are algorithmically estimated values and remain susceptible to real-time traffic fluctuations.

The friction factors (F_{ij}) were not recalibrated using a separate local dataset; rather, the calibrated values provided in the project brief were adopted. Linear interpolation was utilized to derive friction factors for travel times falling between the provided 10-minute increments. Therefore, the accuracy of the trip distribution results depends on how well the project-provided friction factors represent the current travel conditions in the Mirpur area.

For the mode choice analysis, the survey provided individual responses for the three transport alternatives. However, only 11 respondents were included, so the survey results should be treated as a small academic sample rather than a representative sample of Dhaka. In addition, the mode-specific travel time, cost, comfort, safety, and reliability values were obtained for the Mirpur 1–Mirpur 10 trip. These values were therefore used as representative values in the final mode choice calculation, but they may not fully represent the conditions of every O–D pair in the study area. The survey results and detailed supporting data are provided in Appendix E.

3 Model Setup and Detailed Calculations

3.1 Task 1: Trip Generation

The cross-classification method was adopted as the project provides the required household income and land-use data for each Traffic Analysis Zone (TAZ) [1]. Household income categories were used to estimate trip productions, whereas land-use characteristics were employed to estimate trip attractions.

3.1.1 Trip Production

The trip production for each Traffic Analysis Zone (TAZ) was estimated using the trip production rates provided in Table 1 of the project brief. The production rate is based on household income level, where each low-, middle-, and high-income household generates an average of 6, 7, and 9 trips per day, respectively [1]. Therefore, the total trip production for zone i is obtained by summing the contributions from each income group:

$$TP_i = 6L_i + 7M_i + 9H_i \quad (1)$$

where

- TP_i = Total trip production in zone i
- L_i = Number of low-income households in zone i
- M_i = Number of middle-income households in zone i
- H_i = Number of high-income households in zone i

3.1.2 Trip Attraction

Similarly, trip attraction was estimated using the trip attraction rates provided in Table 2 of the project brief. Retail land, non-retail land, and household land attract an average of 6, 4, and 1 trips per unit area (or household unit), respectively. Accordingly, the trip attraction for zone i is calculated as

$$TA_i = 6R_i + 4N_i + H_i \quad (2)$$

where

- TA_i = Total trip attraction in zone i
- R_i = Retail land-use value in zone i
- N_i = Non-retail land-use value in zone i
- H_i = Household land-use value in zone i

A representative calculation for Mirpur 1 is presented below.

$$\begin{aligned} TP_{\text{Mirpur 1}} &= 6(11) + 7(51) + 9(84) \\ &= 66 + 357 + 756 \\ &= 1179 \text{ trips/day} \end{aligned}$$

$$\begin{aligned} TA_{\text{Mirpur 1}} &= 6(100) + 4(88) + 419 \\ &= 600 + 352 + 419 \\ &= 1371 \text{ trips/day} \end{aligned}$$

The detailed input data and supporting trip generation calculations are provided in Appendix A. The calculated trip productions and attractions for all eight TAZs are summarised in Table 1.

Table 1: Raw Trip Production (TP) and Trip Attraction (TA)

Traffic Analysis Zone	Trip Production (TP)	Trip Attraction (TA)
Mirpur 1	1179	1371
Mirpur 10	1908	3063
Mirpur 11	2902	2235
Mirpur DOHS	1558	1083
Pallabi	1664	1119
Kazipara	1885	2632
Shewrapara	1785	1075
Kallyanpur	1326	1856
Total	$\sum TP = 14207$	$\sum TA = 14434$

Since the total trip productions and attractions are not equal, the attraction values were balanced prior to the trip distribution stage. Following standard four-step travel demand modelling practice, trip productions were retained unchanged because they are derived from household socio-economic data and are considered more reliable than attraction estimates.

The production-attraction balancing factor is [5]

$$\begin{aligned} \frac{P}{A} &= \frac{\sum TP}{\sum TA} \\ &= \frac{14207}{14434} \\ &= 0.98427 \end{aligned}$$

Accordingly, the balanced trip attraction for each zone was obtained as

$$TA_i^{\text{bal}} = TA_i \times \frac{\sum TP}{\sum TA} \quad (3)$$

For Mirpur 1,

$$TA_{\text{Mirpur1}}^{\text{bal}} = 1371 \times 0.98427 = 1349$$

The resulting balanced trip attractions are presented in Table 2.

Table 2: Balanced Trip Attraction

Traffic Analysis Zone	Balanced Trip Attraction
Mirpur 1	1349
Mirpur 10	3015
Mirpur 11	2200
Mirpur DOHS	1066
Pallabi	1101
Kazipara	2591
Shewrapara	1058
Kallyanpur	1827
Total	$\sum TA_i^{\text{bal}} = 14207$

Table 3 presents the final balanced trip productions and trip attractions for all Traffic Analysis Zones (TAZs). After applying the balancing factor, the total trip attraction equals the total trip production, thereby satisfying the trip conservation requirement,

$$\sum_{i=1}^n TP_i = \sum_{i=1}^n TA_i^{\text{bal}} = 14\,207 \text{ trips/day.}$$

The balanced trip productions and attractions were subsequently used as inputs to the gravity-based trip distribution model.

Table 3: Final Balanced Trip Production and Trip Attraction

Traffic Analysis Zone	Trip Production (TP_i)	Balanced Trip Attraction (TA_i^{bal})
Mirpur 1	1179	1349
Mirpur 10	1908	3015
Mirpur 11	2902	2200
Mirpur DOHS	1558	1066
Pallabi	1664	1101
Kazipara	1885	2591
Shewrapara	1785	1058
Kallyanpur	1326	1827
Total	$\sum TP_i = 14207$	$\sum TA_i^{\text{bal}} = 14207$

3.2 Task 2: Trip Distribution

3.2.1 Selection of Gravity Model

The gravity model was selected over the growth factor model as it accounts for both trip-end characteristics and spatial separation through travel impedance, enabling future trip distribution predictions based on land-use and network conditions [3].

In contrast, growth factor models primarily adjust an existing O–D matrix using growth factors and are therefore most appropriate when a reliable base-year O–D matrix is available [3]. Since such a base-year O–D matrix was not available for the study area, the growth factor model was considered unsuitable. The project objective also requires estimating travel demand from zonal productions and attractions rather than updating an existing travel pattern. Therefore, the gravity model provides a more appropriate and theoretically sound approach for this study.

3.2.2 Travel Time Data Collection

Travel time between each origin–destination (O–D) pair was obtained using Google Maps. To maintain consistency among all observations, the *Depart at* option was used instead of relying on the instantaneous travel time displayed at the time of data collection. The departure times were fixed at 8:00 AM and 5:00 PM to represent the morning and evening peak periods, respectively. In addition, the *Typical values* option was selected so that the estimated travel times represented typical traffic conditions for the specified departure times rather than a single day-specific traffic condition.

For each O–D pair, travel times were recorded for five weekdays (Sunday–Thursday) at both 8:00 AM and 5:00 PM. Thus, each O–D pair initially provided ten travel-time observations. The morning and evening observations were then averaged for each day to obtain a representative daily travel time. Finally, the daily values were averaged over the five weekdays to obtain the final representative travel time for each O–D pair. The complete Google Maps travel-time matrices used in the analysis are provided in Appendix B.

The final average travel time for an O–D pair was therefore calculated as

$$\bar{T}_{ij} = \frac{1}{5} \sum_{d=1}^5 \left(\frac{T_{ij,d}^{8:00 \text{ AM}} + T_{ij,d}^{5:00 \text{ PM}}}{2} \right) \quad (4)$$

where $\bar{T}_{ij}$ is the final average travel time between origin zone i and destination zone j , and $T_{ij,d}^{8:00 \text{ AM}}$ and $T_{ij,d}^{5:00 \text{ PM}}$ represent the travel-time estimates obtained from Google Maps for weekday d at 8:00 AM and 5:00 PM, respectively.

This procedure was adopted to reduce the influence of variation in travel-time estimates between individual weekdays and to obtain a representative travel time for each O–D pair for use in the subsequent trip distribution analysis.

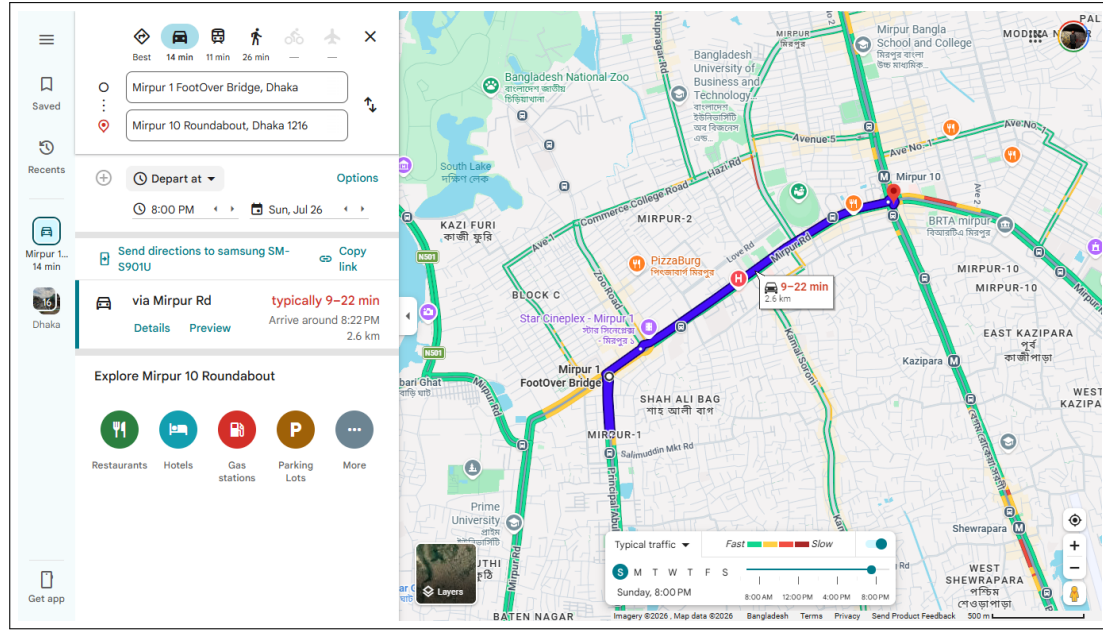


Figure 1: Representative Google Maps Travel Time Observation (Mirpur 1 to Mirpur 10, Sunday, 8:00 AM)

The final average travel time matrix used in the gravity model is presented in Table 4.

Table 4: Representative Average Travel Time Matrix Developed from Google Maps Observations

Origin	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	13	18	33	27	18	22	13
Mirpur 10	12	3	8	19	14	7	12	24
Mirpur 11	16	9	4	15	10	13	17	29
Mirpur DOHS	26	17	13	8	18	22	28	38
Pallabi	27	19	14	10	5	23	27	35
Kazipara	16	11	14	27	21	4	7	28
Shewrapara	21	13	17	32	25	11	6	27
Kallyanpur	17	22	28	41	35	24	25	9

3.2.3 Determination of Friction Factors

The project brief provides calibrated friction factor values corresponding to selected travel times. Since no additional calibrated friction factor dataset specific to the study area was available from reliable sources, the friction factor values provided in the project were adopted without modification to maintain consistency with the intended modelling framework.

Table 5: Travel Time and Corresponding Friction Factors Adopted in This Study

Travel Time (min)	Friction Factor (F)
10	0.72
20	0.52
30	0.41
40	0.35
50	0.26
60	0.17

As observed travel times did not always match the calibrated values in Table 5, intermediate friction factors were estimated via linear interpolation in Microsoft Excel using Equation 5.

$$F = F_1 + \frac{(T - T_1)}{(T_2 - T_1)}(F_2 - F_1) \quad (5)$$

where

- F = interpolated friction factor
- T = observed travel time
- T_1, T_2 = adjacent travel times in Table 5
- F_1, F_2 = corresponding friction factors

For example, the travel time between Mirpur 1 and Mirpur 10 was observed as 13 minutes. From Table 5,

$$\begin{aligned} T_1 &= 10 \text{ min}, & F_1 &= 0.72 \\ T_2 &= 20 \text{ min}, & F_2 &= 0.52 \end{aligned}$$

Therefore,

$$F = 0.72 + \frac{13 - 10}{20 - 10}(0.52 - 0.72) = 0.66$$

Thus, the interpolated friction factor is 0.66.

The complete set of interpolated friction factors used for the O-D pairs is provided in Appendix C.

For observed travel times below the minimum calibrated value of 10 minutes, linear extrapolation was not performed because the resulting friction factors would lie outside the calibrated range. As a modelling assumption, the 10-minute friction factor of 0.72 was therefore assigned to travel times below 10 minutes.

3.2.4 Initial Gravity Model Calculation

The singly constrained gravity model was first applied to estimate the initial O-D matrix. In this model, the total trips produced by each origin are preserved, while the destination totals are determined by the relative attractiveness and travel impedance.

The gravity model estimates the number of trips exchanged between origin zone i and destination zone j as follows [3]

$$T_{ij} = TP_i \left[\frac{TA_j \times F_{ij} \times K_{ij}}{\sum_{j=1}^n TA_j \times F_{ij} \times K_{ij}} \right] \quad (6)$$

where

- T_{ij} = trips from origin i to destination j
- TP_i = trip production of origin zone i
- TA_j = trip attraction of destination zone j
- F_{ij} = friction factor

- K_{ij} = socio-economic adjustment factor

The parameter K_{ij} represents an adjustment factor accounting for socio-economic interactions that are not explicitly represented by trip productions, trip attractions, or travel impedance. Since no observed Origin–Destination survey data were available for calibrating these adjustment factors, it was assumed that

$$K_{ij} = 1$$

for all origin–destination pairs. This assumption is commonly adopted in gravity model applications when calibration data are unavailable, implying that no additional adjustment is applied beyond the effects of productions, attractions, and travel impedance.

For example, the number of trips from Mirpur 1 to Mirpur 10 is calculated as

$$T_{1-2} = \frac{TP_1 \times TA_2 \times F_{1-2}}{\sum_j TA_j \times F_{1-j}}$$

Substituting the corresponding values,

$$T_{1-2} = \frac{1179 \times 3015 \times 0.66}{(1349 \times 0.72) + (3015 \times 0.66) + \dots + (1058 \times 0.498) + (1827 \times 0.66)} = 283$$

The detailed gravity model calculations for all O–D pairs are provided in Appendix D.

Table 6: Initial Origin–Destination Matrix Obtained from the Singly Constrained Gravity Model

Origin	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur	$\sum_j T_{ij}$	TP_i
Mirpur 1	138	283	175	60	69	207	75	172	1179	1179
Mirpur 10	186	440	321	117	143	378	146	177	1908	1908
Mirpur 11	258	691	505	211	252	545	195	245	2902	2902
Mirpur DOHS	126	358	297	157	126	264	94	136	1558	1558
Pallabi	130	356	308	168	173	276	102	152	1664	1664
Kazipara	174	453	302	101	120	401	164	170	1885	1885
Shewrapara	148	429	275	92	111	391	164	175	1785	1785
Kallyanpur	147	282	179	68	79	232	92	247	1326	1326
$\sum_i T_{ij}$	1307	3292	2361	974	1073	2694	1032	1474	14207	
TA_j^{bal}	1349	3015	2200	1066	1101	2591	1058	1827		14207

The data were shared among the students listed in Table 7 because of the large amount of calculation involved.

Table 7: Students Who Shared the Calculation Data

Name	Id
Md. Abeer Azad Niloy	2023-3-22-007
Md. Mohi Uddin	2023-3-22-050
Farhan Labib Jahin	2023-3-22-052
Md. Khairul Islam Hafiz	2023-3-22-058
Tasnim Tabassum Jhoty	2023-3-22-086

3.2.5 Doubly Constrained Gravity Model

The initial singly constrained gravity model preserves the trip production of each origin zone; however, the resulting destination totals do not reproduce the balanced trip attractions obtained in Task 1, as shown in Table 6. The maximum deviation of the calculated trip attractions from the target values is approximately 19.43%. Therefore, the initial O–D matrix requires further balancing.

According to the gravity model calibration procedure, the doubly constrained gravity model is appropriate when the zonal trip productions and attractions are considered more reliable than the available friction factors [5]. In the present study, the trip productions and balanced trip attractions were obtained from the trip generation procedure and are therefore treated as reliable constraints. In contrast, the friction factors were adopted from the project brief and were not calibrated specifically for the present study area. Therefore, the doubly constrained gravity model was selected to adjust the O–D distribution while satisfying both the production and attraction constraints [3, 6].

The attraction values are adjusted based on the ratio between the desired attraction and the actual attraction obtained from the previous iteration. The adjusted attraction for destination zone j at iteration k is calculated as [5]

$$TA_{jk} = TA_{j(k-1)} \left(\frac{TA_j^{\text{bal}}}{C_{j(k-1)}} \right) \quad (7)$$

where

- TA_{jk} = adjusted attraction for destination zone j at iteration k ,
- TA_j^{bal} = desired (target) balanced attraction for destination zone j ,
- $C_{j(k-1)}$ = actual attraction total obtained for destination zone j in the previous iteration,
- $TA_{j(k-1)}$ = attraction value used in the previous iteration,
- j = destination zone number, $j = 1, 2, \dots, n$,
- k = iteration number.

For the first iteration, the attraction values used in the gravity model are the balanced trip attractions obtained in Task 1. Therefore,

$$TA_{j1} = TA_j^{\text{bal}}.$$

The first singly constrained gravity model produced the actual attraction totals shown in Table 6. These values are used as C_{j1} to obtain the adjusted attraction values for the next iteration.

Calculation of Adjusted Trip Attractions

For example, for Mirpur 1, the desired balanced attraction is 1349 trips/day, while the actual attraction obtained from the first gravity model is 1307 trips/day. Therefore, the adjusted attraction for the second iteration is

$$\begin{aligned} TA_{\text{Mirpur 1},2} &= 1349 \times \frac{1349}{1307} \\ &= 1392.35 \end{aligned}$$

The adjusted attraction values were retained at full precision in the subsequent Excel calculations. The values shown in Table 8 are rounded to the nearest whole trip only for presentation.

The same adjustment procedure was applied to all eight destination zones. The detailed calculations are provided in Appendix D.

Table 8: Trip Productions and Adjusted Trip Attractions for the Second Iteration

Traffic Analysis Zone	TP_i	TA_j^{bal}	C_{j1}	TA_{j2}
Mirpur 1	1179	1349	1307	1392
Mirpur 10	1908	3015	3294	2760
Mirpur 11	2902	2200	2362	2049
Mirpur DOHS	1558	1066	973	1168
Pallabi	1664	1101	1074	1129
Kazipara	1885	2591	2693	2493
Shewrapara	1785	1058	1033	1084
Kallyanpur	1326	1827	1471	2268
Total	14207	14207	14207	14343

It should be noted that the sum of the adjusted attraction values, TA_{j2} , is not required to equal the total trip production at this intermediate stage. These adjusted values serve as inputs to the next gravity model iteration, from which a new set of calculated attraction totals is obtained and compared with the target balanced attractions.

Second Iteration of Trip Distribution

Using the adjusted attraction values from Table 8, the gravity model was applied again to obtain a new O–D matrix. The trip production values and friction factors remained unchanged, while the adjusted attraction values were used in place of the original balanced attractions.

Thus, for the second iteration,

$$T_{ij}^{(2)} = TP_i \left[\frac{TA_{j2}F_{ij}}{\sum_{j=1}^n TA_{j2}F_{ij}} \right] \quad (8)$$

For example, for the O–D pair Mirpur 1 to Mirpur 1, the travel time is 6 minutes and the corresponding friction factor is 0.72. Therefore,

$$\begin{aligned} T_{1-1}^{(2)} &= 1179 \times \frac{1392 \times 0.72}{(1392 \times 0.72) + (2760 \times 0.66) + (2049 \times 0.56) + \dots + (2268 \times 0.66)} \\ &= 141 \text{ trips/day.} \end{aligned}$$

The detailed gravity model calculations for all O–D pairs are provided in Appendix D.

Table 9: Final Origin–Destination Matrix from the Doubly Constrained Gravity Model

Origin	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur	$\sum_j T_{ij}$	TP_i
Mirpur 1	141	257	162	65	70	197	76	211	1179	1179
Mirpur 10	193	405	300	128	147	365	150	220	1908	1908
Mirpur 11	268	636	472	232	260	527	201	306	2902	2902
Mirpur DOHS	130	329	278	173	130	255	96	168	1558	1558
Pallabi	135	325	286	183	177	265	105	188	1664	1664
Kazipara	180	417	283	112	124	388	169	212	1885	1885
Shewrapara	154	395	257	101	114	378	169	218	1785	1785
Kallyanpur	148	252	163	73	79	218	93	300	1326	1326
$\sum_i T_{ij}$	1348	3016	2200	1067	1101	2593	1059	1823	14207	
TA_j^{bal}	1349	3015	2200	1066	1101	2591	1058	1827		14207

Note: The O–D trip values in the table are rounded to the nearest whole trip for presentation. The convergence analysis was performed using the unrounded values generated in Microsoft Excel.

Table 10: Convergence Check of the Final Doubly Constrained Gravity Model

Traffic Analysis Zone	Target TA_j	Calculated TA_j	Difference (%)	Status
Mirpur 1	1349	1348.13	0.064	Pass
Mirpur 10	3015	3015.90	0.030	Pass
Mirpur 11	2200	2201.23	0.056	Pass
Mirpur DOHS	1066	1066.59	0.055	Pass
Pallabi	1101	1101.66	0.059	Pass
Kazipara	2591	2592.23	0.048	Pass
Shewrapara	1058	1058.50	0.048	Pass
Kallyanpur	1827	1822.76	0.232	Pass
Maximum Error			0.232	Pass

The final convergence check is presented in Table 10. Since the maximum percentage deviation is 0.232%, which is less than the specified convergence criterion of 5%, the iterative procedure is considered converged. The resulting O–D matrix is therefore adopted as the final trip distribution matrix.

3.3 Task 3: Mode Choice Analysis

3.3.1 Discrete Choice Modelling

Mode choice is the third stage of the traditional four-step travel demand modelling framework. After estimating the number of trips generated and distributed between the Traffic Analysis Zones (TAZs), the next step is to determine how these trips are divided among the available transportation modes. In the present study, the three alternatives specified in the project brief are Bicycle/Rickshaw (BR), Public Transit (PT), and Private Vehicle (PV) [1].

A Discrete Choice Model (DCM) was selected for the mode choice analysis. Discrete choice models represent the decision of an individual traveller among a finite set of mutually exclusive alternatives. The approach is based on the assumption that travellers select the alternative that provides the highest utility. In transport applications, the utility of an alternative is generally represented by a systematic component that can be observed by the analyst and a random component that represents unobserved factors [4, 7].

The general random utility representation is

$$U_{in} = V_{in} + \varepsilon_{in} \quad (9)$$

where

- U_{in} = total utility of alternative i for traveller n ,
- V_{in} = systematic or observable component of utility,
- ε_{in} = random or unobservable component of utility.

In this study, the systematic component is represented by travel time, travel cost, comfort, safety, reliability, and an alternative-specific constant. The DCM approach is appropriate because the mode choice is made at the individual traveller level, while the final predicted mode shares are subsequently applied to the aggregate trip distribution.

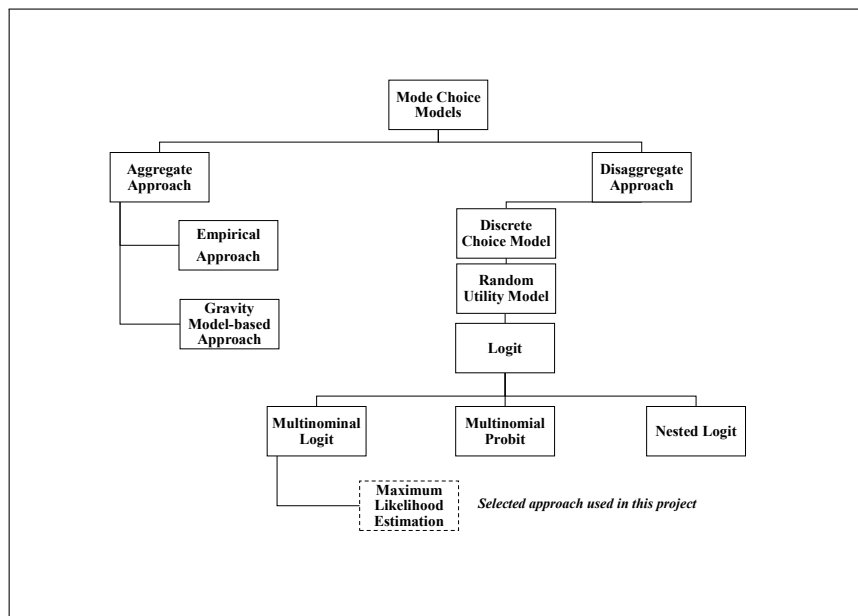


Figure 2: Hierarchy of mode choice modelling approaches and the method adopted in the present study.

3.3.2 Multinomial Logit Model

Among the available discrete choice models, the Multinomial Logit (MNL) model was selected for this study. The MNL model has been widely used in transportation mode choice analysis. Rahman, Fujiwara, and Zhang developed MNL and nested logit models specifically for commuting trips in Dhaka City. Their MNL model used total travel time and total travel cost as deterministic variables together with alternative-specific constants for the available transport modes [4].

For K available alternatives, the probability that traveller n selects alternative i under the MNL model is

$$P_{in} = \frac{\exp(U_{in})}{\sum_{j=1}^K \exp(U_{jn})} \quad (10)$$

where P_{in} is the probability of choosing alternative i and U_{in} is the utility of alternative i for traveller n .

The MNL model is suitable for the present project because only three alternatives are considered and the objective is to estimate the probability of choosing BR, PT, and PV. It also provides a direct framework for calculating the final mode shares.

3.3.3 Maximum Likelihood Estimation

The coefficients of the utility function cannot be directly observed. Therefore, the unknown coefficients were estimated using the Maximum Likelihood Estimation (MLE) method. MLE determines the set of coefficients that makes the observed choices in the survey most likely.

For each respondent n , the probability corresponding to the mode actually chosen by that respondent is used to construct the likelihood. Therefore,

$$L_n = P_{in} \quad (11)$$

where i represents the mode actually chosen by respondent n .

The total likelihood is

$$L = \prod_{n=1}^N P_{in} \quad (12)$$

To simplify the numerical calculation, the logarithm of the likelihood is maximized instead of the likelihood itself. The log-likelihood function is

$$LL = \sum_{n=1}^N \ln(P_{in}) \quad (13)$$

The unknown utility coefficients were estimated in Microsoft Excel using the Solver tool. The objective cell was the total log-likelihood, which was maximized by changing the unknown coefficients. The GRG Nonlinear solving method was used because the MNL probability equation contains exponential terms.

The use of survey data to calibrate utility-function coefficients through maximum likelihood estimation is also demonstrated in standard transportation modelling literature, where observed choices and attributes such as travel time are used to estimate the utility coefficient [2].

3.3.4 Specification of Utility Functions

Following the utility-function structure discussed in the course lecture, the systematic utility of a transportation mode was specified as a function of travel time, travel cost, and additional mode-quality attributes.

The general utility function adopted for the present study is

$$U_x = ASC_x + \beta_{TT} TT_x + \beta_{COST} COST_x + \beta_{COMFORT} COMFORT_x + \beta_{SAFETY} SAFETY_x + \beta_{RELIABILITY} RELIABILITY_x \quad (14)$$

where

- U_x = utility of mode x ,
- ASC_x = alternative-specific constant for mode x ,
- TT_x = total travel time for mode x (min),
- $COST_x$ = travel cost for mode x (BDT),
- $COMFORT_x$ = comfort rating of mode x ,
- $SAFETY_x$ = safety rating of mode x ,
- $RELIABILITY_x$ = reliability rating of mode x ,
- β_{TT} = travel-time coefficient,
- β_{COST} = travel-cost coefficient,
- $\beta_{COMFORT}$ = comfort coefficient,
- β_{SAFETY} = safety coefficient,
- $\beta_{RELIABILITY}$ = reliability coefficient.

The use of total travel time and total travel cost is directly supported by the Dhaka mode-choice model developed by Rahman, Fujiwara, and Zhang. Their estimated coefficients were negative, indicating that longer travel time and higher travel cost reduce the utility of a mode [4].

The additional variables of comfort, safety, and reliability were included because transport service-quality research identifies these as important factors affecting users' perceptions and transportation decisions. In particular, research on bus passengers in Dhaka identified comfort and perceived safety as important service-quality attributes [8]. Reliability and safety have also been identified as important mode-related factors in Dhaka transport research [9].

3.3.5 Literature-Based Coefficients

The coefficients for total travel time, total travel cost, and the alternative-specific constants were adopted from the Dhaka MNL model developed by Rahman, Fujiwara, and Zhang. Their model estimated

$$\beta_{TT} = -0.09496$$

and

$$\beta_{COST} = -0.00813$$

For the alternative-specific constants, private vehicle was used as the base alternative. Therefore,

$$ASC_{PV} = 0,$$

while

$$ASC_{PT} = 1.77996$$

and

$$ASC_{BR} = 2.18344.$$

These values are taken from the MNL estimation results reported for commuting trips in Dhaka City [4].

The literature-based coefficients are summarized in Table 11.

Table 11: Literature-based coefficients adopted for the mode choice model

Parameter	Symbol	Value	Source
Total travel time	β_{TT}	-0.09496	Rahman et al. (2008)
Total travel cost	β_{COST}	-0.00813	Rahman et al. (2008)
BR constant	ASC_{BR}	2.18344	Rahman et al. (2008)
PT constant	ASC_{PT}	1.77996	Rahman et al. (2008)
PV constant	ASC_{PV}	0	Base alternative

It should be noted that the original Dhaka MNL model used total travel time and total travel cost as its main deterministic variables. Therefore, the coefficients for comfort, safety, and reliability were not directly borrowed from that model. Instead, these additional coefficients were estimated using the present survey data.

3.3.6 Survey Data Collection

A concise survey was conducted to obtain the additional mode-specific attributes required for the proposed utility functions. A total of 11 respondents were considered in the survey. Each respondent was asked to identify the transportation mode used or preferred among the three alternatives:

1. Bicycle/Rickshaw (BR),
2. Public Transit (PT),
3. Private Vehicle (PV).

For each respondent, the following attributes were recorded for all three alternatives:

- total travel time (min) (IVTT + OVTT),
- travel cost (BDT),
- comfort rating,
- safety rating,
- reliability rating.

The comfort, safety, and reliability attributes were recorded on a five-point scale. A higher score represents a more favourable perception of the corresponding attribute.

The survey structure was intentionally kept concise because the purpose of the survey in this academic project was to demonstrate the calibration of additional utility coefficients using observed individual-level data. The survey data were then entered into Microsoft Excel, where the MNL probabilities and log-likelihood values were calculated.

3.3.7 Limitations of the Survey and Mode Choice Model

There are some limitations in the mode choice analysis. First, only 11 respondents were included in the survey. Therefore, the estimated coefficients may not represent the actual preferences of the whole population of Dhaka. A larger number of respondents would give more reliable results.

Second, the comfort, safety, and reliability ratings are based on the personal opinions of the respondents. Different people may give different ratings for the same transport mode. Therefore, these values may change if a larger and more diverse group of respondents is surveyed. Previous research also shows that people's perceptions of transport service quality can vary depending on the transport system and local context [8].

Another limitation is related to the travel time and cost data used in the mode choice model. The values of travel time, travel cost, and the alternative-specific constants were taken from a previous Dhaka study rather than being estimated from the present survey. The previous study developed a mode choice model using a much larger dataset of commuting trips in Dhaka City [4]. Therefore, these coefficients may not exactly represent the current travel behaviour in the study area.

Finally, the mode-specific values of travel time, cost, comfort, safety, and reliability collected from the survey were based only on the Mirpur 1–Mirpur 10 trip. These values were then used to calculate mode choice probabilities for the trips obtained from the trip distribution stage. However, travel time, cost, and people's perceptions of comfort, safety, and reliability can vary from one O–D pair to another. Therefore, the same values may not be fully applicable to all O–D pairs in the study area. A more detailed survey covering different O–D pairs would provide more suitable values for a wider travel demand analysis.

3.3.8 MLE Calibration of Unknown Coefficients

The coefficients of comfort, safety, and reliability were initially set to zero:

$$\beta_{COMFORT} = \beta_{SAFETY} = \beta_{RELIABILITY} = 0$$

These values were used only as initial values for the Excel Solver. The Solver was then instructed to maximize the total log-likelihood by changing the three unknown coefficients.

To avoid numerical overflow during the exponential calculation, the maximum utility among the three alternatives was subtracted from each utility before calculating the exponential term. Thus, for respondent n ,

$$U_{\max,n} = \max(U_{BR,n}, U_{PT,n}, U_{PV,n})$$

and

$$E_{in} = \exp(U_{in} - U_{\max,n})$$

The MNL probabilities were then calculated as

$$P_{in} = \frac{E_{in}}{E_{BR,n} + E_{PT,n} + E_{PV,n}}$$

For each respondent, the probability corresponding to the observed choice was selected as the chosen probability. The individual log-likelihood was then calculated as

$$LL_n = \ln(P_{chosen,n})$$

The total log-likelihood is therefore

$$LL = \sum_{n=1}^{11} LL_n$$

For the Solver calibration, the unknown coefficients were constrained within the range

$$0 \leq \beta_{COMFORT}, \beta_{SAFETY}, \beta_{RELIABILITY} \leq 2.$$

The constraints were used as a practical numerical restriction for the small survey-based calibration and should not be interpreted as theoretical limits of the coefficients.

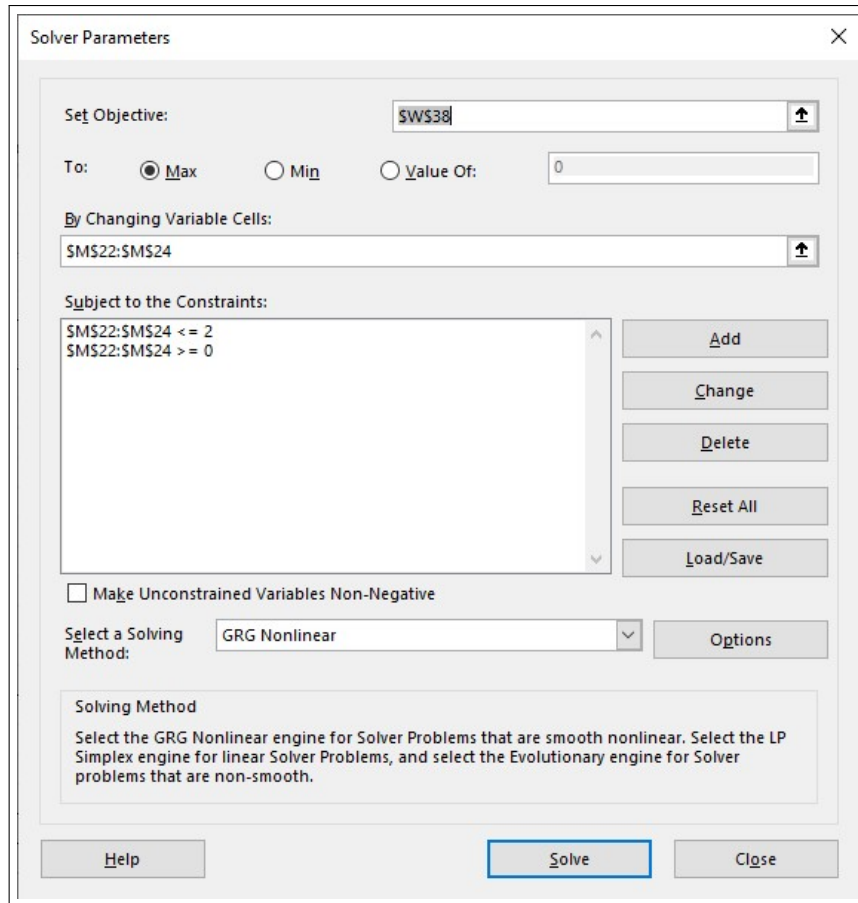


Figure 3: Setup of Solver Parameter

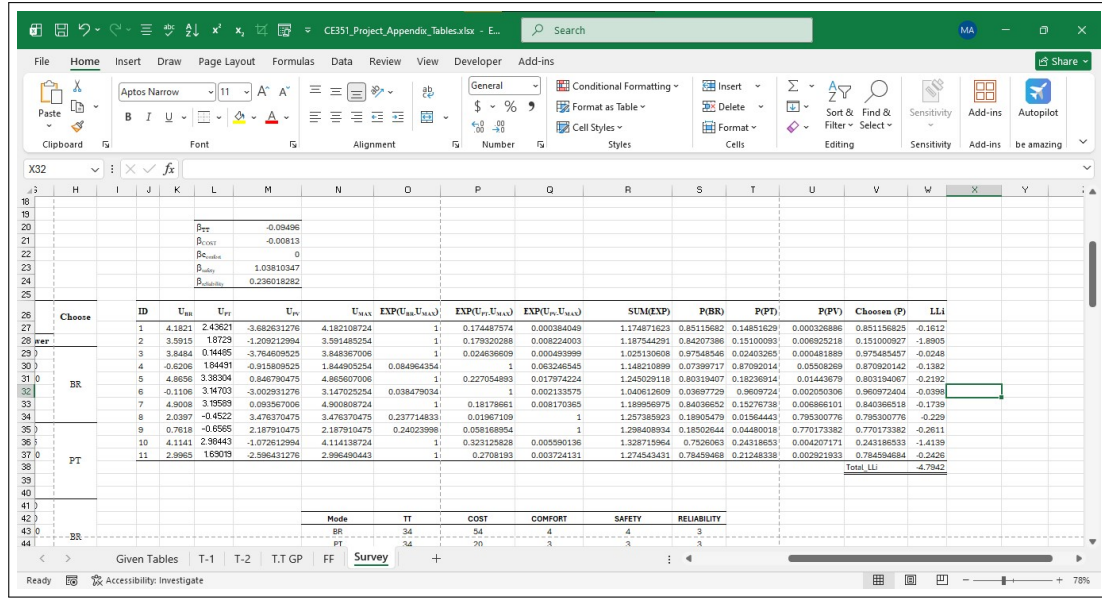


Figure 4: Snapshot of Excel Sheet While Solving MLE

The detailed calculation of MLE are attached to the Appendix E.

3.3.9 MLE Results

The maximum likelihood estimation produced the following coefficients:

Table 12: Estimated coefficients from maximum likelihood estimation

Parameter	Symbol	Estimated coefficient
Travel time	β_{TT}	-0.09496
Travel cost	β_{COST}	-0.00813
Comfort	$\beta_{COMFORT}$	0.00000
Safety	β_{SAFETY}	1.038
Reliability	$\beta_{RELIABILITY}$	0.236

The positive coefficient of safety indicates that, within the survey sample, an increase in the perceived safety rating increases the systematic utility of a mode. Similarly, the positive reliability coefficient indicates that a more reliable mode receives higher utility.

The estimated comfort coefficient is equal to zero. This result does not mean that comfort is unimportant in Dhaka in general. Rather, it indicates that the present 11-person sample did not provide sufficient evidence for a positive comfort coefficient under the specified MNL structure.

The final total log-likelihood obtained from the Excel Solver calculation was

$$LL = -4.794$$

The detailed respondent-level MNL calculation obtained from Microsoft Excel is presented in Table 13.

Table 13: Respondent-level MNL and MLE calculation

ID	U_{BR}	U_{PT}	U_{PV}	P_{BR}	P_{PT}	P_{PV}	P_{chosen}	LL_i	Choice	$U_{\max}$	$\sum E$
1	4.182	2.436	-3.683	0.8512	0.1485	0.0003	0.8512	-0.1612	BR	4.182	1.1749
2	3.591	1.873	-1.209	0.8421	0.1510	0.0069	0.1510	-1.8905	PT	3.591	1.1875
3	3.848	0.145	-3.765	0.9755	0.0240	0.0005	0.9755	-0.0248	BR	3.848	1.0251
4	-0.621	1.845	-0.916	0.0740	0.8709	0.0551	0.8709	-0.1382	PT	1.845	1.1482
5	4.866	3.383	0.847	0.8032	0.1824	0.0144	0.8032	-0.2192	BR	4.866	1.2450
6	-0.111	3.147	-3.003	0.0370	0.9610	0.0021	0.9610	-0.0398	PT	3.147	1.0406
7	4.901	3.196	0.094	0.8404	0.1528	0.0069	0.8404	-0.1739	BR	4.901	1.1900
8	2.040	-0.452	3.476	0.1891	0.0156	0.7953	0.7953	-0.2290	PV	3.476	1.2574
9	0.762	-0.656	2.188	0.1850	0.0448	0.7702	0.7702	-0.2611	PV	2.188	1.2984
10	4.114	2.984	-1.073	0.7526	0.2432	0.0042	0.2432	-1.4139	PT	4.114	1.3287
11	2.996	1.690	-2.596	0.7846	0.2125	0.0029	0.7846	-0.2426	BR	2.996	1.2745
Total log-likelihood								-4.794			

The detailed calculation shows that the three predicted probabilities for each respondent sum to approximately one:

$$P_{BR,n} + P_{PT,n} + P_{PV,n} \approx 1.$$

The chosen probability is the probability corresponding to the mode actually reported by the respondent. The log-likelihood contribution of each respondent is obtained from the natural logarithm of this chosen probability.

3.3.10 Final Utility Functions

After estimating the unknown coefficients, the final utility functions for the three alternatives are obtained by substituting the estimated coefficients into Equation 14.

For Bicycle/Rickshaw (BR),

$$U_{BR} = 2.18344 - 0.09496TT_{BR} - 0.00813COST_{BR} + 1.038SAFETY_{BR} + 0.236RELIABILITY_{BR} \quad (15)$$

Since the estimated comfort coefficient is zero, the comfort term does not appear in the final equation.

For Public Transit (PT),

$$U_{PT} = 1.77996 - 0.09496TT_{PT} - 0.00813COST_{PT} + 1.038SAFETY_{PT} + 0.236RELIABILITY_{PT} \quad (16)$$

For Private Vehicle (PV), the alternative-specific constant is zero because PV is the base alternative:

$$U_{PV} = -0.09496TT_{PV} - 0.00813COST_{PV} + 1.038SAFETY_{PV} + 0.236RELIABILITY_{PV} \quad (17)$$

These utility functions provide the basis for the subsequent calculation of mode choice probabilities and the final mode distribution percentages for the trips obtained from the trip distribution stage.

3.3.11 Determination of Utility Value

After obtaining the final utility functions for BR, PT, and PV, the average values of travel time, travel cost, comfort, safety, and reliability were calculated from the responses of the 11 respondents.

Table 14: Average Mode-Specific Values from the Survey

Mode	TT	COST	COMFORT	SAFETY	RELIABILITY
BR	34	54	4	4	3
PT	34	20	3	3	3
PV	41	316	5	5	4

The representative values of travel time (TT), travel cost (COST), comfort, safety, and reliability obtained from the 11 respondents were substituted into the final utility functions. The resulting utility values were

$$U_{BR} = 2.18344 - 0.09496 \times 34 - 0.00813 \times 54 + 1.038 \times 4 + 0.236 \times 3 = 3.37578$$

$$U_{PT} = 1.77996 - 0.09496 \times 34 - 0.00813 \times 20 + 1.038 \times 3 + 0.236 \times 3 = 2.21072$$

$$U_{PV} = -0.09496 \times 41 - 0.00813 \times 316 + 1.038 \times 5 + 0.236 \times 4 = 7.45828$$

3.3.12 Determination of Mode Choice Probabilities

These utility values were then used in the MNL probability expression presented previously. The resulting mode choice probabilities are calculated as follows.

For Bicycle/Rickshaw (BR),

$$P_{BR} = \frac{e^{3.37578}}{e^{3.37578} + e^{2.21072} + e^{7.45828}} = 0.01650$$

Therefore,

$$P_{BR} = 1.65\%$$

For Public Transit (PT),

$$P_{PT} = \frac{e^{2.21072}}{e^{3.37578} + e^{2.21072} + e^{7.45828}} = 0.00515$$

Therefore,

$$P_{PT} = 0.51\%$$

For Private Vehicle (PV),

$$P_{PV} = \frac{e^{7.45828}}{e^{3.37578} + e^{2.21072} + e^{7.45828}} = 0.97835$$

Therefore,

$$P_{PV} = 97.84\%$$

The calculated probabilities sum to approximately 100%, confirming that the complete trip demand is distributed among the three available transportation modes.

Table 15: Estimated Mode Choice Probabilities

Mode	Utility	Mode Share (%)
Bicycle/Rickshaw (BR)	3.37578	1.65
Public Transit (PT)	2.21072	0.51
Private Vehicle (PV)	7.45828	97.84
Total	–	100.00

3.3.13 Determination of Mode-Specific Trips

The final O–D matrix shows that the study movement between Mirpur 1 and Mirpur DOHS consists of trips in both directions. The number of trips from Mirpur 1 to Mirpur DOHS is 65 trips/day, while the number of trips from Mirpur DOHS to Mirpur 1 is 130 trips/day [Table 9].

Since the available survey-based mode attributes were obtained for the Mirpur 1–Mirpur 10 movement, the resulting mode probabilities are used as representative probabilities for the Mirpur 1–Mirpur DOHS movement. This is a modelling assumption and may not fully represent the actual mode choice for this O–D pair.

The mode-specific trips for each direction were obtained by multiplying the corresponding O–D trips by the mode choice probabilities calculated in the previous section.

For the Mirpur 1 → Mirpur DOHS movement,

$$T_{BR}^{1 \rightarrow DOHS} = 65(0.016500) = 1.07$$

$$T_{PT}^{1 \rightarrow DOHS} = 65(0.005146) = 0.33$$

$$T_{PV}^{1 \rightarrow DOHS} = 65(0.978353) = 63.59$$

Similarly, for the Mirpur DOHS → Mirpur 1 movement,

$$T_{BR}^{DOHS \rightarrow 1} = 130(0.016500) = 2.15$$

$$T_{PT}^{DOHS \rightarrow 1} = 130(0.005146) = 0.67$$

$$T_{PV}^{DOHS \rightarrow 1} = 130(0.978353) = 127.19$$

The total number of trips in both directions is therefore

$$T_{\text{total}} = 65 + 130 = 195 \text{ trips/day}$$

The total number of trips by each mode is obtained by combining the two directions:

$$T_{BR} = 1.07 + 2.15 = 3.22 \text{ trips/day}$$

$$T_{PT} = 0.33 + 0.67 = 1.00 \text{ trips/day}$$

$$T_{PV} = 63.59 + 127.19 = 190.78 \text{ trips/day}$$

Thus, the estimated mode-specific trips for the complete Mirpur 1–Mirpur DOHS movement are summarized in Table 16.

Table 16: Mode-Specific Trips Between Mirpur 1 and Mirpur DOHS

Mode	Mirpur 1 → DOHS	DOHS → Mirpur 1	Total Trips/day
Bicycle/Rickshaw (BR)	1.07	2.15	3.22
Public Transit (PT)	0.33	0.67	1.00
Private Vehicle (PV)	63.59	127.19	190.78
Total	65.00	130.00	195.00

The trip values obtained from the model are treated as estimated or expected trips rather than actual individual trips. Therefore, decimal values can occur after applying the calculated trip proportions to the O–D matrix. The unrounded values were retained in the calculations to avoid introducing rounding errors and to maintain consistency with the total trip values. Rounding was applied only when whole-trip values were required for presentation.

Therefore, out of the total 195 trips/day between Mirpur 1 and Mirpur DOHS in both directions, approximately 3.22 trips/day are estimated to use Bicycle/Rickshaw, 1.00 trip/day to use Public Transit, and 190.78 trips/day to use Private Vehicle. Figure 5 shows the estimated number of trips by transport mode

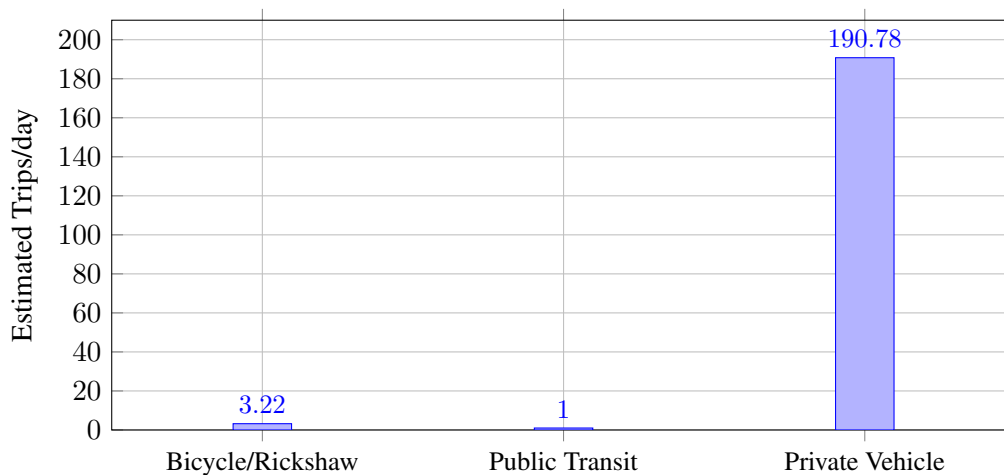


Figure 5: Estimated Mode-Specific Trips Between Mirpur 1 and Mirpur DOHS

for the Mirpur 1–Mirpur DOHS movement. The figure highlights the strong dominance of private-vehicle trips in the model results.

4 Results and Discussion

The main concern of this study is the large difference between the reported commuting time of approximately 55 minutes between Mirpur DOHS and Mirpur 1 and the expected travel time of about 35 minutes under efficient traffic conditions. The results of the first three stages of the four-step travel demand model provide useful evidence for understanding the travel demand behind this congestion problem.

4.1 Trip Generation and Distribution

The trip generation analysis estimated a total of 14,207 trips/day for the eight TAZs. After balancing, the total trip production and attraction were equal, providing a consistent basis for the trip distribution model. The results also show that travel demand is unevenly distributed among the zones. For example, Mirpur 11 generated the highest number of trips, while Mirpur 10 had the highest trip attraction. This indicates that travel demand is strongly influenced by the different residential and activity characteristics of the zones rather than being uniformly distributed across the study area.

The singly constrained gravity model initially produced a maximum deviation of about 19.43% from the target attractions. The doubly constrained model was therefore more appropriate because both trip productions and attractions were treated as important constraints. After iteration, the maximum deviation was reduced to only 0.232%, which is below the 5% convergence criterion. The final O–D matrix was therefore considered acceptable for representing the estimated spatial distribution of trips.

An important result for the project problem is the directional imbalance between Mirpur 1 and Mirpur DOHS. The final matrix estimates 65 trips/day from Mirpur 1 to Mirpur DOHS and 130 trips/day in

the opposite direction. Thus, the corridor carries a considerable amount of travel demand, with the DOHS-to-Mirpur 1 movement being twice as large as the reverse movement. This also demonstrates why the two directions should not be assumed to have equal travel demand when assessing congestion along the corridor.

4.2 Mode Choice and Implications for Congestion

The mode choice analysis produced a predicted mode share of 97.84% for Private Vehicle (PV), compared with 1.65% for Bicycle/Rickshaw (BR) and 0.51% for Public Transit (PT). The very high private-vehicle share is the most important finding in relation to the congestion problem identified in the project proposal.

For the Mirpur 1–Mirpur DOHS movement, the 195 estimated trips/day are distributed as approximately 190.78 private-vehicle trips, 3.22 Bicycle/Rickshaw trips, and 1.00 Public Transit trip. Therefore, the model suggests that most of the estimated travel demand on this movement is concentrated in private vehicles rather than being distributed among different transport modes.

This concentration of demand is a potential contributor to recurrent congestion. A large number of private-vehicle trips requires substantially more road space than a comparable number of trips carried by higher-capacity public transport. Therefore, the high private-vehicle share may contribute to congestion along the corridor. However, the present analysis cannot quantify how much this factor contributes to the reported 55-minute commuting time. However, the present analysis does not establish that private vehicles are the sole cause of the 55-minute travel time. Congestion is also affected by road capacity, intersection operations, traffic control, parking activity, and other network conditions that are outside the scope of the first three stages of this study.

The updated travel-time observations also provide an important comparison with the original problem. The updated Google Maps observations give an average travel time of 26 minutes for the Mirpur DOHS–Mirpur 1 direction, which is below the approximately 35-minute efficient-condition value stated in the proposal. Therefore, the current Google Maps estimate does not reproduce the reported 55-minute commuting experience. The difference may be related to actual traffic conditions and other factors not captured by the typical travel-time values used in this study. This suggests that the expected travel time itself is reasonable; the larger 55-minute commuting experience is more likely associated with traffic conditions rather than simply the physical distance between the two zones.

Nevertheless, the mode-choice result should be interpreted with caution. The utility functions were estimated from a limited survey sample, and the predicted 97.84% private-vehicle share represents the outcome of the calibrated model rather than a direct observation of the entire study area. Thus, the result is most useful for identifying the dominant travel behaviour under the assumptions of the model.

Overall, the analysis indicates that the congestion problem is associated with both the amount and composition of travel demand. The study area generates substantial daily travel, the demand is unevenly distributed between zones, and the selected mode choice model predicts a strong dependence on private vehicles. These findings provide a useful basis for future transport planning, particularly for measures that can improve the attractiveness and reliability of alternative modes and reduce excessive dependence on private vehicles.

5 Conclusion and Recommendations

5.1 Conclusion

This study applied the first three stages of the traditional four-step travel demand modelling framework to the selected Mirpur study area. The analysis provided an estimate of the travel demand generated within the eight TAZs, distributed the trips between the zones using a doubly constrained gravity model, and estimated the choice among Bicycle/Rickshaw, Public Transit, and Private Vehicle.

The results indicate that travel demand is not evenly distributed among the zones and that the movement between Mirpur DOHS and Mirpur 1 is also directional. The mode choice analysis further indicates a strong dependence on private vehicles within the study area. This suggests that the composition of travel demand may be an important factor in the congestion problem. However, the model results alone cannot determine the exact cause of the reported commuting delay because road capacity, intersection performance, parking activity, and traffic control were not included in this study.

The study also has some limitations. The mode choice model was estimated using only 11 survey respondents, and the survey-based mode attributes were mainly obtained for the Mirpur 1–Mirpur 10 movement. Therefore, the estimated mode shares should be considered as model-based estimates rather than direct measurements of the actual travel behaviour of the whole study area. A larger and more representative survey would improve the reliability of the mode choice results.

Overall, the study provides a useful preliminary picture of travel demand, trip distribution, and mode choice in the selected Mirpur area. These results can support further analysis of the Mirpur DOHS–Mirpur 1 corridor, particularly through traffic assignment and detailed investigation of road capacity and intersection conditions.

5.2 Recommendations

Based on the empirical findings and methodological limitations of this study, the following interventions are recommended for future transport planning and research:

1. **Improve public transport service:** Public transport should be made more reliable, comfortable, safe, and competitive with private vehicles.
2. **Reduce dependence on private vehicles:** Transport planning should consider measures that can reduce unnecessary private-vehicle use, particularly along the Mirpur DOHS–Mirpur 1 corridor.
3. **Collect more mode-choice data:** A larger survey covering more respondents and different O–D pairs should be conducted. Travel time, cost, comfort, safety, and reliability should be collected for different movements.
4. **Calibrate the model with local data:** Future studies should use observed O–D and mode-specific travel data to calibrate the friction factors and mode-choice coefficients.
5. **Carry out traffic assignment and corridor analysis:** A further study should examine traffic volumes, road capacity, intersections, delays, and queues along the Mirpur DOHS–Mirpur 1 corridor to better identify the causes of the observed travel-time difference.

References

- [1] East West University, Department of Civil Engineering, “CE351 Transportation Engineering I: Term Project,” 2026, summer 2026, Section 1.
- [2] N. J. Garber and L. A. Hoel, *Traffic and Highway Engineering*, 5th ed. Stamford, CT: Cengage Learning, 2015.
- [3] J. de Dios Ortúzar and L. G. Willumsen, *Modelling Transport*, 4th ed. John Wiley & Sons, 2011.
- [4] M. S. Rahman, A. Fujiwara, and J. Zhang, “Development of travel mode choice models for commuting trips in dhaka city,” in *Proceedings of the International Conference on Sustainable Transport for Developing Countries (STDC)*. Dhaka, Bangladesh: Department of Civil Engineering, BUET, 2008, pp. 57–62.
- [5] F. Y. Huda, “CE351: Transportation Engineering I - Lecture Notes,” 2026, East West University, Department of Civil Engineering.
- [6] F. L. Mannering, S. S. Washburn, and W. P. Kilareski, *Principles of Highway Engineering and Traffic Analysis*, 7th ed. John Wiley & Sons, 2021.
- [7] K. E. Train, *Discrete Choice Methods with Simulation*. Cambridge, UK: Cambridge University Press, 2002.
- [8] M. Quddus, F. Rahman, F. Monsuur, J. de Ona, and M. Enoch, “Analyzing bus passengers’ satisfaction in dhaka using discrete choice models,” *Transportation Research Record*, vol. 2673, no. 2, pp. 758–768, 2019. [Online]. Available: <https://doi.org/10.1177/0361198119825846>
- [9] H. Mohiuddin, M. M. R. Bhuiya, M. M. U. Hasan, and H.-T. Jamme, “How individual perceptions of transportation systems influence mode choice for mobility-challenged people: A case study in dhaka using an integrated choice and latent variable model,” *Transport Policy*, vol. 147, pp. 259–270, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0967070X23003451>

A Trip Generation and Trip Attraction Data

This appendix presents the input data and supporting calculations used for Task 1 (Trip Generation). The appendix includes the trip-rate data adopted from the project brief, household income data, land-use data, raw trip production and attraction values, balanced trip attractions, and the final balanced trip production and attraction values.

A.1 Trip Production and Trip Attraction Rates

Household Income Group	Average Trips/day
Low-income	6
Middle-income	7
High-income	9

Land-Use Category	Average Trips/day
Retail	6
Non-retail	4
Household	1

A.2 Household Income Data

Traffic Analysis Zone	Low-income	Middle-income	High-income
Mirpur 1	11	51	84
Mirpur 10	10	57	161
Mirpur 11	15	106	230
Mirpur DOHS	15	85	97
Pallabi	12	59	131
Kazipara	12	61	154
Shewrapara	15	75	130
Kallyanpur	16	60	90

A.3 Land-Use Data

Traffic Analysis Zone	Retail	Non-retail	Household
Mirpur 1	100	88	419
Mirpur 10	362	139	335
Mirpur 11	82	241	779
Mirpur DOHS	72	89	295
Pallabi	66	156	99
Kazipara	78	515	104
Shewrapara	102	99	67
Kallyanpur	139	228	110

A.4 Raw Trip Production and Trip Attraction

Traffic Analysis Zone	Trip Production (TP_i)	Trip Attraction (TA_i)
Mirpur 1	1179	1371
Mirpur 10	1908	3063
Mirpur 11	2902	2235
Mirpur DOHS	1558	1083
Pallabi	1664	1119
Kazipara	1885	2632
Shewrapara	1785	1075
Kallyanpur	1326	1856
Total	14207	14434

A.5 Balanced Trip Attraction

Traffic Analysis Zone	Balanced Trip Attraction (TA_i^{bal})
Mirpur 1	1349
Mirpur 10	3015
Mirpur 11	2200
Mirpur DOHS	1066
Pallabi	1101
Kazipara	2591
Shewrapara	1058
Kallyanpur	1827
Total	14207

A.6 Final Balanced Trip Production and Trip Attraction

Traffic Analysis Zone	Trip Production (TP_i)	Balanced Trip Attraction (TA_i^{bal})
Mirpur 1	1179	1349
Mirpur 10	1908	3015
Mirpur 11	2902	2200
Mirpur DOHS	1558	1066
Pallabi	1664	1101
Kazipara	1885	2591
Shewrapara	1785	1058
Kallyanpur	1326	1827
Total	14207	14207

Note: The balanced attraction values shown in the main report are rounded to the nearest whole trip for presentation. The underlying calculations may retain full numerical precision.

B Google Maps Travel Time Data

B.1 Sunday, 8 AM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	16	21	36	29	19	25	12
Mirpur 10	12	2	8	15	16	4	9	25
Mirpur 11	20	13	5	13	9	18	19	33
Mirpur DOHS	22	16	13	7	18	19	25	34
Pallabi	22	19	17	11	6	24	28	29
Kazipara	11	7	11	25	19	3	5	25
Shewrapara	16	10	14	27	21	9	5	19
Kallyanpur	16	20	25	39	32	19	19	8

B.2 Sunday, 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	18	16	32	26	17	21	12
Mirpur 10	10	2	6	15	14	4	9	20
Mirpur 11	16	10	4	14	8	14	18	29
Mirpur DOHS	26	16	15	8	21	20	26	37
Pallabi	28	20	15	9	5	24	28	33
Kazipara	15	7	13	25	14	3	5	26
Shewrapara	19	11	17	32	25	10	5	23
Kallyanpur	17	25	29	43	36	22	25	9

B.3 Monday, 8 AM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	11	17	42	25	22	28	16
Mirpur 10	29	3	15	27	6	29	37	48
Mirpur 11	16	18	5	16	10	16	17	31
Mirpur DOHS	26	19	11	6	19	30	32	42
Pallabi	24	17	10	12	5	23	25	44
Kazipara	22	36	19	36	28	8	15	38
Shewrapara	30	18	23	43	33	15	8	46
Kallyanpur	15	24	29	44	38	35	38	8

B.4 Monday, 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	11	16	31	25	23	28	17
Mirpur 10	10	4	9	24	18	8	10	26
Mirpur 11	15	8	4	16	10	15	23	31
Mirpur DOHS	25	18	11	6	19	30	33	41
Pallabi	25	16	10	12	6	23	25	40
Kazipara	22	13	19	35	28	5	9	38
Shewrapara	31	18	25	45	33	15	8	40
Kallyanpur	15	24	29	44	38	36	42	8

B.5 Tuesday, 8 AM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	13	21	28	26	15	19	11
Mirpur 10	9	2	6	16	13	3	8	18
Mirpur 11	13	4	2	13	10	7	11	21
Mirpur DOHS	25	16	13	7	15	18	25	33
Pallabi	25	16	13	6	3	19	24	30
Kazipara	14	6	12	20	18	3	5	22
Shewrapara	17	10	14	25	21	8	4	19
Kallyanpur	16	19	26	34	33	18	19	8

B.6 Tuesday, 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	12	18	29	25	17	19	12
Mirpur 10	10	2	7	16	13	4	9	21
Mirpur 11	15	5	2	13	10	10	14	26
Mirpur DOHS	28	17	14	7	14	22	28	40
Pallabi	29	20	15	6	3	24	28	33
Kazipara	15	8	13	22	20	3	5	25
Shewrapara	20	11	16	26	25	10	5	26
Kallyanpur	17	25	29	40	36	22	25	9

B.7 Wednesday, 8 AM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	4	7	17	29	25	16	18	12
Mirpur 10	9	2	7	19	14	3	8	20
Mirpur 11	12	4	2	12	11	8	12	25
Mirpur DOHS	25	16	12	6	14	18	25	37
Pallabi	26	18	14	8	4	20	27	32
Kazipara	13	6	12	25	18	3	5	25
Shewrapara	16	10	15	27	21	9	5	21
Kallyanpur	16	19	26	39	32	19	19	8

B.8 Wednesday, 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	13	18	33	26	17	19	11
Mirpur 10	10	2	7	19	13	4	9	20
Mirpur 11	16	5	3	16	10	10	14	26
Mirpur DOHS	29	17	13	9	20	22	26	40
Pallabi	32	20	15	9	5	25	28	34
Kazipara	16	7	13	27	20	3	5	28
Shewrapara	20	11	17	32	25	10	5	24
Kallyanpur	17	23	30	42	35	25	22	9

B.9 Thursday, 8 AM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	12	16	31	25	16	20	12
Mirpur 10	9	2	6	19	15	3	9	20
Mirpur 11	15	10	5	14	9	14	19	29
Mirpur DOHS	22	15	13	7	15	18	25	33
Pallabi	25	17	15	9	5	21	27	33
Kazipara	13	6	10	24	18	3	5	25
Shewrapara	16	10	14	27	21	9	5	22
Kallyanpur	16	20	25	39	33	19	19	8

B.10 Thursday, 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	12	16	31	28	16	20	12
Mirpur 10	9	2	6	19	14	3	9	20
Mirpur 11	15	10	5	14	9	14	19	29
Mirpur DOHS	22	15	13	7	15	18	25	33
Pallabi	25	17	15	9	5	21	27	33
Kazipara	13	6	10	24	18	3	5	25
Shewrapara	16	10	14	27	21	9	5	22
Kallyanpur	16	20	25	39	33	19	19	8

B.11 Sunday, Average of 8 AM and 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	17	19	34	28	18	23	12
Mirpur 10	11	2	7	15	15	4	9	23
Mirpur 11	18	12	5	14	9	16	19	31
Mirpur DOHS	24	16	14	8	20	20	26	36
Pallabi	25	20	16	10	6	24	28	31
Kazipara	13	7	12	25	17	3	5	26
Shewrapara	18	11	16	30	23	10	5	21
Kallyanpur	17	23	27	41	34	21	22	9

B.12 Monday, Average of 8 AM and 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	11	17	37	25	23	28	17
Mirpur 10	20	4	12	26	12	19	24	37
Mirpur 11	16	13	5	16	10	16	20	31
Mirpur DOHS	26	19	11	6	19	30	33	42
Pallabi	25	17	10	12	6	23	25	42
Kazipara	22	25	19	36	28	7	12	38
Shewrapara	31	18	24	44	33	15	8	43
Kallyanpur	15	24	29	44	38	36	40	8

B.13 Tuesday, Average of 8 AM and 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	13	20	29	26	16	19	12
Mirpur 10	10	2	7	16	13	4	9	20
Mirpur 11	14	5	2	13	10	9	13	24
Mirpur DOHS	27	17	14	7	15	20	27	37
Pallabi	27	18	14	6	3	22	26	32
Kazipara	15	7	13	21	19	3	5	24
Shewrapara	19	11	15	26	23	9	5	23
Kallyanpur	17	22	28	37	35	20	22	9

B.14 Wednesday, Average of 8 AM and 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	5	10	18	31	26	17	19	12
Mirpur 10	10	2	7	19	14	4	9	20
Mirpur 11	14	5	3	14	11	9	13	26
Mirpur DOHS	27	17	13	8	17	20	26	39
Pallabi	29	19	15	9	5	23	28	33
Kazipara	15	7	13	26	19	3	5	27
Shewrapara	18	11	16	30	23	10	5	23
Kallyanpur	17	21	28	41	34	22	21	9

B.15 Thursday, Average of 8 AM and 5 PM

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	12	16	31	27	16	20	12
Mirpur 10	9	2	6	19	15	3	9	20
Mirpur 11	15	10	5	14	9	14	19	29
Mirpur DOHS	22	15	13	7	15	18	25	33
Pallabi	25	17	15	9	5	21	27	33
Kazipara	13	6	10	24	18	3	5	25
Shewrapara	16	10	14	27	21	9	5	22
Kallyanpur	16	20	25	39	33	19	19	8

B.16 Final, Average of 5 Days

Zone	Mirpur 1	Mirpur 10	Mirpur 11	Mirpur DOHS	Pallabi	Kazipara	Shewrapara	Kallyanpur
Mirpur 1	6	13	18	33	27	18	22	13
Mirpur 10	12	3	8	19	14	7	12	24
Mirpur 11	16	9	4	15	10	13	17	29
Mirpur DOHS	26	17	13	8	18	22	28	38
Pallabi	27	19	14	10	5	23	27	35
Kazipara	16	11	14	27	21	4	7	28
Shewrapara	21	13	17	32	25	11	6	27
Kallyanpur	17	22	28	41	35	24	25	9

C Friction Factor Calculation

C.1 Project-Provided Friction Factors

Travel Time	Friction Factor
10	0.72
20	0.52
30	0.41
40	0.35
50	0.26
60	0.17

C.2 Interpolated Friction Factors

Travel Time	Friction Factor	Travel Time	Friction Factor	Travel Time	Friction Factor	Travel Time	Friction Factor
< 10	0.72	22	0.498	35	0.38	48	0.278
10	0.72	23	0.487	36	0.374	49	0.269
11	0.7	24	0.476	37	0.368	50	0.26
12	0.68	25	0.465	38	0.362	51	0.251
13	0.66	26	0.454	39	0.356	52	0.242
14	0.64	27	0.443	40	0.35	53	0.233
15	0.62	28	0.432	41	0.341	54	0.224
16	0.6	29	0.421	42	0.332	55	0.215
17	0.58	30	0.41	43	0.323	56	0.206
18	0.56	31	0.404	44	0.314	57	0.197
19	0.54	32	0.398	45	0.305	58	0.188
20	0.52	33	0.392	46	0.296	59	0.179
21	0.509	34	0.386	47	0.287	60	0.17

D Gravity Model Calculation

D.1 Initial Gravity Model Calculation

$i - j$	TP_i	TA_j	F_{ij}	$TP_iTA_jF_{ij}$	TA_jF_{ij}	$\sum_j TA_jF_{ij}$	T_{ij}
Mirpur 1–Mirpur 1	1179	1349	0.72	1145139.12	971.28	8282.459	138
Mirpur 1–Mirpur 10	1179	3015	0.66	2346092.10	1989.90		283
Mirpur 1–Mirpur 11	1179	2200	0.56	1452528.00	1232.00		175
Mirpur 1–Mirpur DOHS	1179	1066	0.392	492671.088	417.872		59
Mirpur 1–Pallabi	1179	1101	0.443	575048.997	487.743		69
Mirpur 1–Kazipara	1179	2591	0.56	1710681.84	1450.96		207
Mirpur 1–Shewrapara	1179	1058	0.498	621196.236	526.884		75
Mirpur 1–Kallyanpur	1179	1827	0.66	1421661.78	1205.82		172
Mirpur 10–Mirpur 1	1908	1349	0.68	1750246.56	917.32	9407.012	186
Mirpur 10–Mirpur 10	1908	3015	0.72	4141886.40	2170.80		440
Mirpur 10–Mirpur 11	1908	2200	0.72	3022272	1584		321
Mirpur 10–Mirpur DOHS	1908	1066	0.54	1098321.12	575.64		117
Mirpur 10–Pallabi	1908	1101	0.64	1344453.12	704.64		143
Mirpur 10–Kazipara	1908	2591	0.72	3559412.16	1865.52		378
Mirpur 10–Shewrapara	1908	1058	0.68	1372691.52	719.44		146
Mirpur 10–Kallyanpur	1908	1827	0.476	1659296.016	869.652		176
Mirpur 11–Mirpur 1	2902	1349	0.6	2348878.8	809.4	9110.707	258
Mirpur 11–Mirpur 10	2902	3015	0.72	6299661.6	2170.8		691
Mirpur 11–Mirpur 11	2902	2200	0.72	4596768	1584		505
Mirpur 11–Mirpur DOHS	2902	1066	0.62	1917989.84	660.92		211
Mirpur 11–Pallabi	2902	1101	0.72	2300473.44	792.72		253
Mirpur 11–Kazipara	2902	2591	0.66	4962594.12	1710.06		545
Mirpur 11–Shewrapara	2902	1058	0.58	1780783.28	613.64		195
Mirpur 11–Kallyanpur	2902	1827	0.421	2232122.634	769.167		245
DOHS–Mirpur 1	1558	1349	0.454	954190.868	612.446	7605.974	125
DOHS–Mirpur 10	1558	3015	0.58	2724474.6	1748.7		358
DOHS–Mirpur 11	1558	2200	0.66	2262216	1452		297
DOHS–Mirpur DOHS	1558	1066	0.72	1195796.16	767.52		157
DOHS–Pallabi	1558	1101	0.56	960600.48	616.56		126
DOHS–Kazipara	1558	2591	0.498	2010315.444	1290.318		264
DOHS–Shewrapara	1558	1058	0.432	712093.248	457.056		94
DOHS–Kallyanpur	1558	1827	0.362	1030420.692	661.374		135
Pallabi–Mirpur 1	1664	1349	0.443	994418.048	597.607	7618.718	131
Pallabi–Mirpur 10	1664	3015	0.54	2709158.4	1628.1		356
Pallabi–Mirpur 11	1664	2200	0.64	2342912	1408		308
Pallabi–Mirpur DOHS	1664	1066	0.72	1277153.28	767.52		168
Pallabi–Pallabi	1664	1101	0.72	1319086.08	792.72		173
Pallabi–Kazipara	1664	2591	0.487	2099663.488	1261.817		276
Pallabi–Shewrapara	1664	1058	0.443	779906.816	468.694		102
Pallabi–Kallyanpur	1664	1827	0.38	1155248.64	694.26		152
Kazipara–Mirpur 1	1885	1349	0.6	1525719	809.4	8777.091	174
Kazipara–Mirpur 10	1885	3015	0.7	3978292.5	2110.5		453

Continued on next page

continued from previous page

$i - j$	TP_i	TA_j	F_{ij}	$TP_iTA_jF_{ij}$	TA_jF_{ij}	$\sum_j TA_jF_{ij}$	T_{ij}
Kazipara– Mirpur 11	1885	2200	0.64	2654080	1408		302
Kazipara– Mirpur DOHS	1885	1066	0.443	890168.63	472.238		101
Kazipara– Pallabi	1885	1101	0.509	1056370.965	560.409		120
Kazipara– Kazipara	1885	2591	0.72	3516505.2	1865.52		401
Kazipara– Shewrapara	1885	1058	0.72	1435917.6	761.76		164
Kazipara– Kallyanpur	1885	1827	0.432	1487762.64	789.264		170
Shewrapara– Mirpur 1	1785	1349	0.509	1225654.185	686.641	8273.595	148
Shewrapara– Mirpur 10	1785	3015	0.66	3551971.5	1989.9		429
Shewrapara– Mirpur 11	1785	2200	0.58	2277660	1276		275
Shewrapara– Mirpur DOHS	1785	1066	0.398	757318.38	424.268		92
Shewrapara– Pallabi	1785	1101	0.465	913857.525	511.965		110
Shewrapara– Kazipara	1785	2591	0.7	3237454.5	1813.7		391
Shewrapara– Shewrapara	1785	1058	0.72	1359741.6	761.76		164
Shewrapara– Kallyanpur	1785	1827	0.443	1444709.385	809.361		175
Kallyanpur– Mirpur 1	1326	1349	0.58	1037488.92	782.42	7056.902	147
Kallyanpur– Mirpur 10	1326	3015	0.498	1990949.22	1501.47		282
Kallyanpur– Mirpur 11	1326	2200	0.432	1260230.4	950.4		179
Kallyanpur– Mirpur DOHS	1326	1066	0.341	482008.956	363.506		68
Kallyanpur– Pallabi	1326	1101	0.38	554771.88	418.38		79
Kallyanpur– Kazipara	1326	2591	0.476	1635377.016	1233.316		232
Kallyanpur– Shewrapara	1326	1058	0.465	652352.22	491.97		92
Kallyanpur– Kallyanpur	1326	1827	0.72	1744273.44	1315.44		247

D.2 Second Iteration

$i - j$	TP_i	TA_j	F_{ij}	$TP_iTA_jF_{ij}$	TA_jF_{ij}	$\sum_j TA_jF_{ij}$	T_{ij}
Mirpur 1- Mirpur 1	1179	1392	0.72	1181849.861	1002.417185	8362.215586	141
Mirpur 1- Mirpur 10	1179	2760	0.66	2147698.756	1821.627444		257
Mirpur 1- Mirpur 11	1179	2049	0.56	1352672.391	1147.304827		162
Mirpur 1- Mirpur DOHS	1179	1168	0.392	539833.9201	457.8744021		65
Mirpur 1- Pallabi	1179	1129	0.443	589664.9895	500.1399402		71
Mirpur 1- Kazipara	1179	2493	0.56	1645761.204	1395.895847		197
Mirpur 1- Shewrapara	1179	1084	0.498	636377.0314	539.7599927		76
Mirpur 1- Kallyanpur	1179	2268	0.66	1765194.022	1497.195948		211
Mirpur 10- Mirpur 1	1908	1392	0.68	1806355.767	946.7273413	9373.899411	193
Mirpur 10- Mirpur 10	1908	2760	0.72	3791634.723	1987.229939		404
Mirpur 10- Mirpur 11	1908	2049	0.72	2814502.642	1475.106206		300
Mirpur 10- Mirpur DOHS	1908	1168	0.54	1203462.128	630.7453499		128
Mirpur 10- Pallabi	1908	1129	0.64	1378625.02	722.5498008		147
Mirpur 10- Kazipara	1908	2493	0.72	3424331.927	1794.723232		365
Mirpur 10- Shewrapara	1908	1084	0.68	1406237.359	737.0216768		150
Mirpur 10- Kallyanpur	1908	2268	0.476	2060250.511	1079.795865		220
Mirpur 11- Mirpur 1	2902	1392	0.6	2424178.892	835.3476541	9063.570066	267
Mirpur 11- Mirpur 10	2902	2760	0.72	5766941.282	1987.229939		636
Mirpur 11- Mirpur 11	2902	2049	0.72	4280758.211	1475.106206		472
Mirpur 11- Mirpur DOHS	2902	1168	0.62	2101596.784	724.1891054		232
Mirpur 11- Pallabi	2902	1129	0.72	2358944.462	812.8685259		260
Mirpur 11- Kazipara	2902	2493	0.66	4774262.919	1645.162963		527
Mirpur 11- Shewrapara	2902	1084	0.58	1824302.067	628.6361361		201
Mirpur 11- Kallyanpur	2902	2268	0.421	2771495.715	955.0295364		306
DOHS- Mirpur 1	1558	1392	0.454	984780.2115	632.0797249	7589.074471	130
DOHS- Mirpur 10	1558	2760	0.58	2494083.975	1600.824117		329
DOHS- Mirpur 11	1558	2049	0.66	2106697.514	1352.180689		278
DOHS- Mirpur DOHS	1558	1168	0.72	1310268.34	840.9937998		173
DOHS- Pallabi	1558	1129	0.56	985016.0159	632.2310757		130
DOHS- Kazipara	1558	2493	0.498	1934023.667	1241.350236		255
DOHS- Shewrapara	1558	1084	0.432	729495.3849	468.2255359		96
DOHS- Kallyanpur	1558	2268	0.362	1279412.918	821.1892926		169
Pallabi- Mirpur 1	1664	1392	0.443	1026296.99	616.765018	7628.356022	135
Pallabi- Mirpur 10	1664	2760	0.54	2480062.963	1490.422454		325
Pallabi- Mirpur 11	1664	2049	0.64	2181845.98	1311.205517		286
Pallabi- Mirpur DOHS	1664	1168	0.72	1399413.683	840.9937998		183
Pallabi- Pallabi	1664	1129	0.72	1352613.227	812.8685259		177
Pallabi- Kazipara	1664	2493	0.487	2019980.939	1213.930853		265
Pallabi- Shewrapara	1664	1084	0.443	798966.181	480.1479453		105
Pallabi- Kallyanpur	1664	2268	0.38	1434404.457	862.0219093		188
Kazipara- Mirpur 1	1885	1392	0.6	1574630.328	835.3476541	8725.76189	180
Kazipara- Mirpur 10	1885	2760	0.7	3641874.867	1932.029107		417
Kazipara- Mirpur 11	1885	2049	0.64	2471622.399	1311.205517		283
Kazipara- Mirpur DOHS	1885	1168	0.443	975383.441	517.4447963		112

Continued on next page

continued from previous page

$i - j$	TP_i	TA_j	F_{ij}	$TP_iTA_jF_{ij}$	TA_jF_{ij}	$\sum_j TA_jF_{ij}$	T_{ij}
Kazipara- Pallabi	1885	1129	0.509	1083220.695	574.6528884		124
Kazipara- Kazipara	1885	2493	0.72	3383053.293	1794.723232		388
Kazipara- Shewrapara	1885	1084	0.72	1471008.558	780.3758931		169
Kazipara- Kallyanpur	1885	2268	0.432	1847267.582	979.9828022		212
Shewrapara- Mirpur 1	1785	1392	0.509	1264946.069	708.6532599	8238.602747	154
Shewrapara- Mirpur 10	1785	2760	0.66	3251604.987	1821.627444		395
Shewrapara- Mirpur 11	1785	2049	0.58	2121079.799	1188.279999		257
Shewrapara- Mirpur DOHS	1785	1168	0.398	829815.5906	464.8826838		101
Shewrapara- Pallabi	1785	1129	0.465	937084.9975	524.9775896		114
Shewrapara- Kazipara	1785	2493	0.7	3114592.609	1744.869809		378
Shewrapara- Shewrapara	1785	1084	0.72	1392970.969	780.3758931		169
Shewrapara- Kallyanpur	1785	2268	0.443	1793810.881	1004.936068		218
Kallyanpur- Mirpur 1	1326	1392	0.58	1070748.623	807.5027323	7218.194021	148
Kallyanpur- Mirpur 10	1326	2760	0.498	1822587.938	1374.500708		252
Kallyanpur- Mirpur 11	1326	2049	0.432	1173594.498	885.0637238		163
Kallyanpur- Mirpur DOHS	1326	1168	0.341	528151.1146	398.304008		73
Kallyanpur- Pallabi	1326	1129	0.38	568872.49	429.0139442		79
Kallyanpur- Kazipara	1326	2493	0.476	1573314.21	1186.51147		218
Kallyanpur- Shewrapara	1326	1084	0.465	668294.4055	503.9927643		93
Kallyanpur- Kallyanpur	1326	2268	0.72	2165761.993	1633.30467		300

E Mode Choice Modelling Data

E.1 Survey Responses for Mode Choice Analysis

ID	BR		PT		PV		Choose
	Attribute	Answer	Attribute	Answer	Attribute	Answer	
1	IVTT (min)	20	IVTT (min)	20	IVTT (min)	30	BR
	OVTt (min)	5	OVTt (min)	10	OVTt (min)	30	
	Cost (taka)	60	Cost (taka)	10	Cost (taka)	350	
	Comfort (1–5)	3	Comfort (1–5)	1	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	3	Safety (1–5)	4	
	Reliability (1–5)	3	Reliability (1–5)	2	Reliability (1–5)	3	
2	IVTT (min)	15	IVTT (min)	20	IVTT (min)	20	PT
	OVTt (min)	7	OVTt (min)	5	OVTt (min)	25	
	Cost (taka)	40	Cost (taka)	10	Cost (taka)	250	
	Comfort (1–5)	3	Comfort (1–5)	1	Comfort (1–5)	5	
	Safety (1–5)	3	Safety (1–5)	2	Safety (1–5)	4	
	Reliability (1–5)	3	Reliability (1–5)	2	Reliability (1–5)	4	
3	IVTT (min)	25	IVTT (min)	35	IVTT (min)	30	BR
	OVTt (min)	6	OVTt (min)	4	OVTt (min)	40	
	Cost (taka)	60	Cost (taka)	30	Cost (taka)	400	
	Comfort (1–5)	4	Comfort (1–5)	1	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	2	Safety (1–5)	5	
	Reliability (1–5)	4	Reliability (1–5)	1	Reliability (1–5)	4	
4	IVTT (min)	40	IVTT (min)	30	IVTT (min)	25	PT
	OVTt (min)	12	OVTt (min)	7	OVTt (min)	15	
	Cost (taka)	80	Cost (taka)	30	Cost (taka)	400	
	Comfort (1–5)	3	Comfort (1–5)	2	Comfort (1–5)	5	
	Safety (1–5)	2	Safety (1–5)	3	Safety (1–5)	5	
	Reliability (1–5)	3	Reliability (1–5)	3	Reliability (1–5)	4	
5	IVTT (min)	17	IVTT (min)	15	IVTT (min)	10	BR
	OVTt (min)	5	OVTt (min)	10	OVTt (min)	20	
	Cost (taka)	40	Cost (taka)	10	Cost (taka)	300	
	Comfort (1–5)	4	Comfort (1–5)	3	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	3	Safety (1–5)	5	
	Reliability (1–5)	4	Reliability (1–5)	4	Reliability (1–5)	4	
6	IVTT (min)	35	IVTT (min)	15	IVTT (min)	10	PT
	OVTt (min)	10	OVTt (min)	10	OVTt (min)	30	
	Cost (taka)	70	Cost (taka)	10	Cost (taka)	500	
	Comfort (1–5)	3	Comfort (1–5)	2	Comfort (1–5)	5	
	Safety (1–5)	2	Safety (1–5)	3	Safety (1–5)	4	
	Reliability (1–5)	2	Reliability (1–5)	3	Reliability (1–5)	3	
7	IVTT (min)	15	IVTT (min)	15	IVTT (min)	12	BR
	OVTt (min)	5	OVTt (min)	7	OVTt (min)	15	
	Cost (taka)	30	Cost (taka)	10	Cost (taka)	300	
	Comfort (1–5)	4	Comfort (1–5)	3	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	3	Safety (1–5)	4	

Continued on next page

continued from previous page

ID	BR		PT		PV		Choose
	Attribute	Answer	Attribute	Answer	Attribute	Answer	
8	Reliability (1–5)	3	Reliability (1–5)	2	Reliability (1–5)	4	PV
	IVTT (min)	25	IVTT (min)	35	IVTT (min)	20	
	OVTT (min)	10	OVTT (min)	12	OVTT (min)	2	
	Cost (taka)	50	Cost (taka)	10	Cost (taka)	70	
	Comfort (1–5)	3	Comfort (1–5)	2	Comfort (1–5)	5	
	Safety (1–5)	3	Safety (1–5)	2	Safety (1–5)	5	
	Reliability (1–5)	2	Reliability (1–5)	1	Reliability (1–5)	4	
9	IVTT (min)	40	IVTT (min)	50	IVTT (min)	30	PV
	OVTT (min)	13	OVTT (min)	10	OVTT (min)	3	
	Cost (taka)	55	Cost (taka)	40	Cost (taka)	100	
	Comfort (1–5)	3	Comfort (1–5)	3	Comfort (1–5)	5	
	Safety (1–5)	3	Safety (1–5)	3	Safety (1–5)	5	
	Reliability (1–5)	4	Reliability (1–5)	2	Reliability (1–5)	4	
10	IVTT (min)	22	IVTT (min)	20	IVTT (min)	15	PT
	OVTT (min)	5	OVTT (min)	5	OVTT (min)	20	
	Cost (taka)	45	Cost (taka)	30	Cost (taka)	350	
	Comfort (1–5)	3	Comfort (1–5)	3	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	3	Safety (1–5)	4	
	Reliability (1–5)	3	Reliability (1–5)	3	Reliability (1–5)	4	
11	IVTT (min)	25	IVTT (min)	32	IVTT (min)	20	BR
	OVTT (min)	10	OVTT (min)	5	OVTT (min)	20	
	Cost (taka)	60	Cost (taka)	20	Cost (taka)	450	
	Comfort (1–5)	4	Comfort (1–5)	3	Comfort (1–5)	5	
	Safety (1–5)	4	Safety (1–5)	3	Safety (1–5)	4	
	Reliability (1–5)	2	Reliability (1–5)	2	Reliability (1–5)	3	

E.2 Detailed MNL and Maximum Likelihood Estimation Calculation

β_{TT}	-0.09496
β_{COST}	-0.00813
$\beta_{COMFORT}$	0
β_{SAFETY}	1.03810347
$\beta_{RELIABILITY}$	0.236018282

ID	U_{BR}	U_{PT}	U_{PV}	$U_{\max}$	$\exp(U_{BR} - U_{\max})$	$\exp(U_{PT} - U_{\max})$	$\exp(U_{PV} - U_{\max})$	$\sum \exp$	$P(BR)$	$P(PT)$	$P(PV)$	P_{chosen}	LL_i
1	4.1821	2.4362	-3.68263	4.18210872	1	0.174487574	0.000384049	1.1748716	0.8511568	0.14852	0.000326886	0.851156825	-0.1612
2	3.5915	1.8729	-1.20921	3.59148525	1	0.179320288	0.008224003	1.1875443	0.8420739	0.151	0.006925218	0.151000927	-1.8905
3	3.8484	0.1448	-3.76461	3.84836701	1	0.024636609	0.000493999	1.0251306	0.9754855	0.02403	0.000481889	0.975485457	-0.0248
4	-0.621	1.8449	-0.91581	1.84490525	0.084964354	1	0.063246545	1.1482109	0.0739972	0.87092	0.05508269	0.870920142	-0.1382
5	4.8656	3.383	0.84679	4.86560701	1	0.227054893	0.017974224	1.2450291	0.8031941	0.18237	0.01443679	0.803194067	-0.2192
6	-0.111	3.147	-3.00293	3.14702525	0.038479034	1	0.002133575	1.0406126	0.0369773	0.96097	0.002050306	0.960972404	-0.0398
7	4.9008	3.1959	0.09357	4.90080872	1	0.181786610	0.008170365	1.1899570	0.8403665	0.15277	0.006866101	0.840366518	-0.1739
8	2.0397	-0.4522	3.47637	3.47637048	0.237714833	0.019671090	1	1.2573859	0.1890548	0.01564	0.795300776	0.795300776	-0.2290
9	0.7618	-0.6565	2.18791	2.18791048	0.240239980	0.058168954	1	1.2984089	0.1850264	0.0448	0.770173382	0.770173382	-0.2611
10	4.1141	2.9844	-1.07261	4.11413872	1	0.323125828	0.005590136	1.3287160	0.7526063	0.24319	0.004207171	0.243186533	-1.4139
11	2.9965	1.6902	-2.59643	2.99649044	1	0.270819300	0.003724131	1.2745434	0.7845947	0.21248	0.002921933	0.784594684	-0.2426
Total Log-Likelihood													-4.7942